ELECTIVE COURSE SYLLABUS

Class- IDD

**GENERAL**

1.1 TITLE: **MICROPROCESSORS AND MICROCONTROLLERS**

1.2 *COURSE NUMBER (if known): BM311.16

1.3 CREDITS: 3L-0T-0P

1.4 SEMESTER-OFFERED:

1.5 PRE-REQUISITES: Basics of electronics, electrical and programming skill

2. OBJECTIVE: The course aims to provide an understanding of working of microprocessor and design a microcontroller for small applications.

3. COURSE TOPICS:

1. **Introduction:** Microprocessors, Microprocessors Architecture and Microcomputer system, 8085 Microprocessor Architecture and Memory interfacing, 8085 Memory organization, Instruction set, Timing Diagrams, Memory mapping, Types of Data transfer schemes, Interrupts. (**12 lectures**)

2. **Peripheral Devices and Interfacing:** Peripheral interface devices 8155,8255, Programmable interval Timer 8253, Programmable interrupt controller 8259A, The keyboard / Display controller 8279, Programmable communication Interface 8251 USART, DMA controller 8257, Programmable DMA interface 8237. (**10lectures**)

**3. Introduction to Microcontrollers:** Single chip microcontroller, Introduction to 8 bit microcontrollers, Architecture of 8051, Signal description of 8051, Register set of 8051, Instruction set of 8051, Memory and I/O interfacing by 8051, Interrupts of 8051, Application of 8051 microcontroller.(**8lectures**)

**4. Recent Trends in Microprocessor and Embedded Devices:** The16 bit and higher processor, salient feature, system architecture. Embedded devices their features and architecture.(**6 lectures**)

**4. READINGS**

**4.1 TEXTBOOK:**

1. Microprocessor Architecture, Programming and Applications with 8085, Gaonker Ramesh S., Third Edition, Pen ram International, 2000.
2. Intel Microprocessors Architecture, Programming and Interfacing, Roy A.K. and Bhurhandi K.M, McGraw Hill International Edition, 2001.

**4.2 *REFERENCE BOOKS:**

1. Microprocessors and Interfacing, Programming and hardware Second Edition, DoughlasV.Hall, Tata McGraw Hill Edition, 1997.